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United States Patent

12417460

Kind Code

B2

Date of Patent

September 16, 2025

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Systems and methods to enable payments in the absence of a point of sale device

Abstract

A method and system for processing a transaction between a merchant and a consumer using a consumer mobile device wherein the consumer mobile device is in communication with a payment processing computing device having a memory are provided. The method includes receiving merchant data associated with the merchant, receiving a consumer alias associated with the consumer, receiving, a transaction request message from the consumer mobile device, the transaction request message including a merchant identifier, a transaction amount, a device identifier associated with the consumer mobile device, and consumer verification data, retrieving the account number based on the device identifier and the consumer verification data, retrieving the merchant data based on the merchant identifier, generating an authorization request message including the account number, the transaction amount, and the merchant data, and transmitting the authorization request message over a payment network to an issuer bank.

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Appl. No.: 17/317601

Filed: May 11, 2021

Prior Publication Data

Document Identifier

Publication Date

US 20210264422 A1

Aug. 26, 2021

Related U.S. Application Data

Publication Classification

Int. Cl.: **G06Q20/40** (20120101); **G06Q20/12** (20120101); **G06Q20/20** (20120101); **G06Q20/32** (20120101); **G06Q20/38** (20120101)

U.S. Cl.:

CPC **G06Q20/4015** (20200501); **G06Q20/12** (20130101); **G06Q20/20** (20130101);
G06Q20/322 (20130101); **G06Q20/385** (20130101);

Field of Classification Search

CPC: G06Q (20/322); G06Q (20/3223); G06Q (20/325-3255); G06Q (20/385)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims priority to U.S. application Ser. No. 14/043,339 filed on Oct. 1, 2013, which claims priority to and the benefit of the filing date of U.S. Provisional Application No. 61/859,401 filed on Jul. 29, 2013, both of which are hereby incorporated by reference in their entirety.

BACKGROUND

(1) This disclosure relates generally to consumer initiated financial transaction payment requests and, more particularly, to computer systems and computer-based methods for capturing data by a mobile device and processing the data for completing a purchase without a point of sale device.

(2) At least some known merchants engaged in commerce do not use or otherwise have access to

traditional point-of-sale (POS) devices. However, customers of these merchants may be inclined to use non-cash options when making purchases. For example, merchants may gather at impromptu locations such as, but, not limited to farmers' markets, flea markets, rummage sales, stalls, and kiosks, but may not have access to electricity, Internet, or to other networks, and resources.

(3) Accordingly, it would be desirable to provide a system and/or method for a consumer-initiated financial transaction without a POS device.

BRIEF DESCRIPTION

(4) In one embodiment, a computer-implemented method for processing a transaction between a merchant and a consumer using a consumer mobile device wherein the consumer mobile device is in communication with a payment processing computing device having a memory and wherein the method includes receiving merchant data associated with the merchant at the payment processing computing device, receiving a consumer alias associated with payment credentials of the consumer at the payment processing computing device, receiving, by the payment processing computing device, a transaction request message from the consumer mobile device, the transaction request message including a merchant identifier, a transaction amount, a device identifier associated with the consumer mobile device, and consumer verification data, retrieving, from the memory, the account number based on the device identifier and the consumer verification data, retrieving, from the memory, the merchant and associated acquirer bank data based on the merchant identifier, generating, by the payment processing computing device, an authorization request message including the account number, the transaction amount, and the merchant data, transmitting the authorization request message over a payment network to an issuer bank.

(5) In another embodiment, a computer-implemented method for processing a transaction between a consumer and a merchant using a consumer mobile device where the consumer mobile device is in communication with a payment processing computing device having a memory and wherein the method includes acquiring a price for a transaction between the consumer and the merchant using the consumer mobile device, acquiring merchant data associated with the merchant using the consumer mobile device, transmitting a device identifier associated with the consumer mobile device, the acquired price, and the merchant data in a transaction request message to the payment processing computing device, and receiving, by the consumer mobile device, a transaction authorization message from an issuer bank through a payment network, the transaction authorization message including an indication of approval or denial of the transaction request message.

(6) In yet another embodiment, a mobile remote payment system for processing a transaction between a consumer and a merchant without using a point of sale (POS) device includes a payment processing computing device communicatively coupled to an electronic communications network, said payment processing computing device comprising a processor communicatively coupled to a memory device, a consumer mobile device comprising an image capture component and a geolocation component, said consumer mobile device communicatively coupleable to the payment processing computing device through the electronic communications network, and a merchant consumer mobile device communicatively coupleable to the payment processing computing device through the electronic communications network, where the payment processing computing device is programmed to receive merchant data associated with the merchant at the payment processing computing device, receive a consumer alias associated with the consumer at the payment processing computing device, receive, by the payment processing computing device, a transaction request message from the consumer mobile device, the transaction request message including a merchant identifier, a transaction amount, a device identifier associated with the consumer mobile device, and consumer verification data, retrieve, from the memory, the account number based on the device identifier and the consumer verification data, retrieve, from the memory, the merchant data based on the merchant identifier, generate, by the payment processing computing device, an authorization request message including the account number, the transaction amount, and the

merchant and acquirer data, and transmit the authorization request message over a payment network to an issuer bank.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1-6 show example embodiments of the methods and systems described herein.
- (2) FIG. 1A is a schematic diagram illustrating an example multi-party transaction card industry system for enabling ordinary payment-by-card transactions in which merchants and card issuers do not need to have a one-to-one special relationship.
- (3) FIG. 1B is a schematic diagram illustrating another example multi-party payment card industry system using a mobile remote payment subsystem for enabling payment by mobile device transactions in which merchants and card issuers do not necessarily have a one-to-one relationship and merchants do not have a point-of-sale (POS) device.
- (4) FIG. 2 is a simplified block diagram of an example processing system including a plurality of computer devices in accordance with one embodiment of the present disclosure.
- (5) FIG. 3 is an expanded block diagram of an example embodiment of a server architecture of a processing system including other computer devices in accordance with one embodiment of the present disclosure.
- (6) FIG. 4 illustrates an example configuration of a user system operated by a user, such as the cardholder shown in FIG. 1.
- (7) FIG. 5 illustrates an example configuration of a server system such as server system shown in FIGS. 2 and 3.
- (8) FIG. 6 illustrates a component view of the mobile remote payment system illustrated in FIG. 2.

DETAILED DESCRIPTION

- (9) Embodiments of the methods and systems described herein relate to permitting financial transactions using an interchange network in instances where the merchant may not have access to a POS device. Such a transaction may be consumer-initiated in that the consumer acquires data about the transaction with a portable mobile device, such as, a feature phone, smartphone, or tablet and then generates and transmits an authorization request to the interchange network.
- (10) The methods and systems described herein may be implemented using computer programming or engineering techniques including computer software, firmware, hardware or any combination or subset thereof, wherein the technical effect may include at least one of: (a) receiving merchant data associated with the merchant at the payment processing computing device, (b) receiving a consumer alias associated with payment credentials of the consumer at the payment processing computing device, (c) receiving, by the payment processing computing device, a transaction request message from the consumer mobile device, the transaction request message including a merchant identifier, a transaction amount, a device identifier associated with the consumer mobile device, and consumer verification data (d) retrieving, from the memory, the account number based on the device identifier and the consumer verification data, (e) retrieving, from the memory, the merchant data based on the merchant identifier, (f) generating, by the payment processing computing device, an authorization request message including the account number, the transaction amount, the merchant data, and information identifying an acquirer bank associated with the merchant, and (g) transmitting the authorization request message to the acquirer bank and over a payment network to an issuer bank.
- (11) As used herein, the terms “transaction card,” “financial transaction card,” and “payment card” refer to any suitable transaction card, such as a credit card, a debit card, a prepaid card, a charge card, a membership card, a promotional card, a frequent flyer card, an identification card, a prepaid card, a gift card, and/or any other device that may hold payment account information, such as

mobile phones, smartphones, personal digital assistants (PDAs), key fobs, and/or computers. Each type of transactions card can be used as a method of payment for performing a transaction.

(12) In one embodiment, a computer program is provided, and the program is embodied on a computer readable medium. In an example embodiment, the system is executed on a single computer system, without requiring a connection to a sever computer. In a further example embodiment, the system is being run in a Windows® environment (Windows is a registered trademark of Microsoft Corporation, Redmond, Washington). In yet another embodiment, the system is run on a mainframe environment and a UNIX® server environment (UNIX is a registered trademark of AT&T located in New York, New York). The application is flexible and designed to run in various different environments without compromising any major functionality. In some embodiments, the system includes multiple components distributed among a plurality of computing devices. One or more components may be in the form of computer-executable instructions embodied in a computer-readable medium. The systems and processes are not limited to the specific embodiments described herein. In addition, components of each system and each process can be practiced independent and separate from other components and processes described herein. Each component and process can also be used in combination with other assembly packages and processes.

(13) The following detailed description illustrates embodiments of the disclosure by way of example and not by way of limitation. It is contemplated that the disclosure has general application to processing financial transaction data by a third party in industrial, commercial, and residential applications.

(14) As used herein, an element or step recited in the singular and proceeded with the word “a” or “an” should be understood as not excluding plural elements or steps, unless such exclusion is explicitly recited. Furthermore, references to “example embodiment” or “one embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

(15) FIG. 1A is a schematic diagram illustrating an example multi-party transaction card industry system **20** for enabling ordinary payment-by-card transactions in which merchants **24** and card issuers **30** do not need to have a one-to-one special relationship. Embodiments described herein may relate to a transaction card system, such as a payment card network operated by MasterCard International Incorporated, the assignee of the present disclosure. The payment card network, as described herein, is a four-party payment card interchange network that includes a plurality of special purpose processors and data structures stored in one or more memory devices communicatively coupled to the processors, and a set of proprietary communications standards promulgated by MasterCard International Incorporated for the exchange of financial transaction data and the settlement of funds between financial institutions that are members of the payment network.

(16) In a typical transaction card system, a financial institution called the “issuer” issues a transaction card, such as a credit card, to a consumer or cardholder **22**, who uses the transaction card to tender payment for a purchase from a merchant **24**. To accept payment with the transaction card, merchant **24** must normally establish an account with a financial institution that is part of the financial payment system. This financial institution is usually called the “merchant bank,” the “acquiring bank,” or the “acquirer.” When cardholder **22** tenders payment for a purchase with a transaction card, merchant **24** requests authorization from a merchant bank **26** for the amount of the purchase. The request may be performed over the telephone, but is usually performed through the use of a point-of-sale terminal, which reads cardholder's **22** account information from a magnetic stripe, a chip, or embossed characters on the transaction card and communicates electronically with the transaction processing computers of merchant bank **26**. Alternatively, merchant bank **26** may authorize a third party to perform transaction processing on its behalf. In this case, the point-of-sale terminal will be configured to communicate with the third party. Such a third party is usually called

a “merchant processor,” an “acquiring processor,” or a “third party processor.”

(17) Using an interchange network **28**, computers of merchant bank **26** or merchant processor will communicate with computers of an issuer bank **30** to determine whether cardholder's **22** account **32** is in good standing and whether the purchase is covered by cardholder's **22** available credit line. Based on these determinations, the request for authorization will be declined or accepted. If the request is accepted, an authorization code is issued to merchant **24**.

(18) When a request for authorization is accepted, the available credit line of cardholder's **22** account **32** is decreased. Normally, a charge for a payment card transaction is not posted immediately to cardholder's **22** account **32** because bankcard associations, such as the payment card network operated by MasterCard International Incorporated, have promulgated rules that do not allow merchant **24** to charge, or “capture,” a transaction until goods are shipped or services are delivered. However, with respect to at least some debit card transactions, a charge may be posted at the time of the transaction. When merchant **24** ships or delivers the goods or services, merchant **24** captures the transaction by, for example, appropriate data entry procedures on the point-of-sale terminal. This may include bundling of approved transactions daily for standard retail purchases. If cardholder **22** cancels a transaction before it is captured, a “void” is generated. If cardholder **22** returns goods after the transaction has been captured, a “credit” is generated. Interchange network **28** and/or issuer bank **30** stores the transaction card information, such as a type of merchant, amount of purchase, date of purchase, in a database **120** (shown in FIG. 2).

(19) After a purchase has been made, a clearing process occurs to transfer additional transaction data related to the purchase among the parties to the transaction, such as merchant bank **26**, interchange network **28**, and issuer bank **30**. More specifically, during and/or after the clearing process, additional data, such as a time of purchase, a merchant name, a type of merchant, purchase information, cardholder account information, a type of transaction, itinerary information, information regarding the purchased item and/or service, and/or other suitable information, is associated with a transaction and transmitted between parties to the transaction as transaction data, and may be stored by any of the parties to the transaction. In the example embodiment, when cardholder **22** purchases travel, such as airfare, a hotel stay, and/or a rental car, at least partial itinerary information is transmitted during the clearance process as transaction data. When interchange network **28** receives the itinerary information, interchange network **28** routes the itinerary information to database **120**.

(20) After a transaction is authorized and cleared, the transaction is settled among merchant **24**, merchant bank **26**, and issuer bank **30**. Settlement refers to the transfer of financial data or funds among merchant's **24** account, merchant bank **26**, and issuer bank **30** related to the transaction. Usually, transactions are captured and accumulated into a “batch,” which is settled as a group. More specifically, a transaction is typically settled between issuer bank **30** and interchange network **28**, and then between interchange network **28** and merchant bank **26**, and then between merchant bank **26** and merchant **24**.

(21) FIG. 1B is a schematic diagram illustrating another example multi-party payment card industry system using a mobile remote payment subsystem **34** for enabling payment by mobile device transactions in which merchants and card issuers do not necessarily have a one-to-one relationship and merchants do not have a point-of-sale (POS) device.

(22) In various regions of the world, merchants may not have POS devices in their shops or stalls. To facilitate financial transactions in these regions, an alternate system of charging a customers account, clearing the transaction, and crediting a merchant's account other messaging alternatives may be implemented.

(23) In the example embodiment, mobile remote payment system **34** includes a payment processing computing device **36** communicatively coupled to an electronic communications network **38**. Payment processing computing device **36** includes a processor **40** communicatively coupled to a memory device **42**. Mobile remote payment system **34** also includes a consumer mobile device **44**

that includes an image capture component **46** such as, but, not limited to a digital camera, and a geolocation component **48**. Consumer mobile device **44** refers to a mobile computing device used by a cardholder for making a purchase, transmitting a personal record, storing a personal record, and/or displaying a personal record. Consumer mobile device **44** may be embodied in for example, but, not limited to a computer, smartphone, PDA, tablet, or any other device capable of performing the functions described herein. Consumer mobile device **44** may store payment card information and utilize the stored payment card information to perform payment transactions. Consumer mobile device **44** is communicatively couplable to payment processing computing device **36** through electronic communications network **38**.

(24) Mobile remote payment system **34** includes a merchant mobile device **50** communicatively couplable to payment processing computing device **36** through electronic communications network **38**.

(25) During a transaction, a consumer may wish to purchase a product from a merchant in a shop or stall that does not have a POS device. To initiate a financial transaction for the product, the consumer may scan a price of the product, either from a tag associated with the product or even a handwritten price. Alternatively, the price may be entered into a consumer mobile device **44** using a manual input using a user interface (UI) presented on the consumer mobile device or using speech input. The consumer then collects data about the merchant he wishes to do business with. In one embodiment, the consumer uses image capture component **46** to acquire an image of the merchant information such as, but, not limited to a business license or other business identifying information, or an optically recognizable code such as, but, not limited to a barcode or QR code that is able to provide the merchant's information that can be decoded by payment processing computing device **36**. Moreover, the consumer may also use geolocation component **48** to determine a geographical location of the transaction which can be transmitted to payment processing computing device **36** for a determination of the merchant associated with that location. In other embodiments, both image capture component **46** and geolocation component **48** can be used together to more accurately determine the proper merchant for the transaction. When the information is acquired, the consumer indicates the information is accurate and consumer mobile device **44** transmits the information in an authorization request message to payment processing computing device **36** through network **38**. Payment processing computing device **36** communicates with interchange **28** for further processing and/or forwarding to issuer **30**. Payment processing computing device **36** then receives the authorization request response and forwards the response to both consumer mobile device **44** and merchant mobile device **50**.

(26) In various embodiments, payment processing computing device **36** is programmed to receive merchant data associated with the merchant, consumer alias, which can be used to obtain an account number associated with the consumer at the payment processing computing device, and a transaction request message from consumer mobile device **44**, wherein the transaction request message includes a merchant identifier, a transaction amount, a device identifier associated with consumer mobile device **44**, and consumer verification data. Rather than using an account number to identify the consumer a consumer alias may be used, such as, but not limited to, a phone number, email address or other identifier representative of the consumer. Payment processing computing device **36** is also programmed to retrieve, from memory **42**, the consumer account number based on the device identifier, the consumer verification data and the merchant data based on the merchant identifier. In various embodiments, the device identifier and the consumer verification data are passed to interchange **28** where the account number, acquirer data, and merchant data are determined, rather than being determined by payment processing computing device **36**.

(27) Payment processing computing device **36** is further programmed to generate an authorization request message including the account number, the transaction amount, and the merchant data, transmit the authorization request message over a payment network to an issuer bank, receive an authorization response message from the issuer bank through the payment network, and transmit

the authorization response message to consumer mobile device **44** and merchant mobile device **50**. (28) FIG. 2 is a simplified block diagram of an example processing system **100** including a plurality of computer devices in accordance with one embodiment of the present disclosure. In the example embodiment, system **100** may be used for performing payment-by-card transactions and/or mobile remote payments as of part processing the financial transaction. For example, system **100** may receive financial transaction information from mobile remote payment system **34**, such as, but, not limited to a price for a product or service, merchant information, and identifying information for consumer mobile device **44**. Some of the information may be passed on to interchange network **28** for processing or some of the information may be processed locally at payment processing computing device **36** and the resultant processed data may then be transmitted to interchange network **28**. For example, an image of the price may be acquired by consumer mobile device **44** and the image file may be processed using optical character recognition (OCR) onboard consumer mobile device **44** to generate a text or other format file containing the price information. Alternatively, the image file may be transmitted to payment processing computing device **36**, where payment processing computing device **36** would process the image file to extract the price information. In another embodiment, the image file could be transmitted to interchange network **28**, where the price information could be extracted. The processing may be selectable based on the computing power of the respective devices and/or a bandwidth of the communications links between the components.

(29) More specifically, in the example embodiment, system **100** includes a server system **112**, and a plurality of client sub-systems, also referred to as client systems **114**, connected to server system **112**. In one embodiment, client systems **114** are computers including a web browser, such that server system **112** is accessible to client systems **114** using the Internet. Client systems **114** are interconnected to the Internet through many interfaces including a network, such as a local area network (LAN) or a wide area network (WAN), dial-in-connections, cable modems, and special high-speed Integrated Services Digital Network (ISDN) lines. Client systems **114** could be any device capable of interconnecting to the Internet including a web-based phone, PDA, or other web-based connectable equipment.

(30) System **100** also includes point-of-sale (POS) terminals **118**, which may be connected to client systems **114** and may be connected to server system **112**. POS terminals **118** are interconnected to the Internet through many interfaces including a network, such as a local area network (LAN) or a wide area network (WAN), dial-in-connections, cable modems, wireless modems, and special high-speed ISDN lines. POS terminals **118** could be any device capable of interconnecting to the Internet and including an input device capable of reading information from a consumer's financial transaction card.

(31) A database server **116** is connected to database **120**, which contains information on a variety of matters, as described below in greater detail. In one embodiment, centralized database **120** is stored on server system **112** and can be accessed by potential users at one of client systems **114** by logging onto server system **112** through one of client systems **114**. In an alternative embodiment, database **120** is stored remotely from server system **112** and may be non-centralized.

(32) Database **120** may include a single database having separated sections or partitions or may include multiple databases, each being separate from each other. Database **120** may store transaction data generated as part of sales activities conducted over the processing network including data relating to merchants, account holders or customers, issuers, acquirers, purchases made. Database **120** may also store account data including at least one of a cardholder name, a cardholder address, an account number, and other account identifier. Database **120** may also store merchant data including a merchant identifier that identifies each merchant registered to use the network, and instructions for settling transactions including merchant bank account information. Database **120** may also store purchase data associated with items being purchased by a cardholder from a merchant, and authorization request data. Database **120** may store merchant and associated

acquirer data, an account number and alias associated with the consumer, a merchant identifier, a transaction amount, a device identifier associated with the consumer mobile device, and/or consumer verification data for processing according to the method described in the present disclosure.

(33) In the example embodiment, one of client systems **114** may be associated with acquirer bank **26** (shown in FIG. 1) while another one of client systems **114** may be associated with issuer bank **30** (shown in FIG. 1). POS terminal **118** may be associated with a participating merchant **24** (shown in FIG. 1) or may be a computer system and/or mobile system used by a cardholder making an on-line purchase or payment. Server system **112** may be associated with interchange network **28**. In the example embodiment, server system **112** is associated with a network interchange, such as interchange network **28**, and may be referred to as an interchange computer system. Server system **112** may be used for processing transaction data. In addition, client systems **114** and/or POS **118** may include a computer system associated with at least one of an online bank, a bill payment outsourcer, an acquirer bank, an acquirer processor, an issuer bank associated with a transaction card, an issuer processor, a remote payment system, a biller, and/or mobile remote payment system **34**. Mobile remote payment system **34** may be associated with interchange network **28** or with an outside third party in a contractual relationship with interchange network **28**. Accordingly, each party involved in processing transaction data are associated with a computer system shown in system **100** such that the parties can communicate with one another as described herein.

(34) Using the interchange network, the computers of the merchant bank or the merchant processor will communicate with the computers of the issuer bank to determine whether the consumer's account is in good standing and whether the purchase is covered by the consumer's available credit line. Based on these determinations, the request for authorization will be declined or accepted. If the request is accepted, an authorization code is issued to the merchant.

(35) When a request for authorization is accepted, the available credit line of consumer's account is decreased. Normally, a charge is not posted immediately to a consumer's account because bankcard associations, such as the payment card network, have promulgated rules that do not allow a merchant to charge, or "capture," a transaction until goods are shipped or services are delivered. When a merchant ships or delivers the goods or services, the merchant captures the transaction by, for example, appropriate data entry procedures on the point-of-sale terminal. If a consumer cancels a transaction before it is captured, a "void" is generated. If a consumer returns goods after the transaction has been captured, a "credit" is generated.

(36) For debit card transactions, when a request for a PIN authorization is approved by the issuer, the consumer's account is decreased. Normally, a charge is posted immediately to a consumer's account. The bankcard association then transmits the approval to the acquiring processor for distribution of goods/services, or information or cash in the case of an ATM.

(37) After a transaction is captured, the transaction is settled between the merchant, the merchant bank, and the issuer. Settlement refers to the transfer of financial data or funds between the merchant's account, the merchant bank, and the issuer related to the transaction. Usually, transactions are captured and accumulated into a "batch," which is settled as a group.

(38) The financial transaction cards or payment cards discussed herein may include credit cards, debit cards, a charge card, a membership card, a promotional card, prepaid cards, and gift cards. These cards can all be used as a method of payment for performing a transaction. As described herein, the term "financial transaction card" or "payment card" includes cards such as credit cards, debit cards, and prepaid cards, but also includes any other devices that may hold payment account information, such as mobile phones, personal digital assistants (PDAs), key fobs, or other devices, etc.

(39) FIG. 3 is an expanded block diagram of an example embodiment of a server architecture of a processing system **122** including other computer devices in accordance with one embodiment of the present disclosure. Components in system **122**, identical to components of system **100** (shown

in FIG. 2), are identified in FIG. 3 using the same reference numerals as used in FIG. 2. System **122** includes server system **112**, client systems **114**, and POS terminals **118**. Server system **112** further includes database server **116**, a transaction server **124**, a web server **126**, a fax server **128**, a directory server **130**, and a mail server **132**. A storage device **134** is coupled to database server **116** and directory server **130**. Servers **116**, **124**, **126**, **128**, **130**, and **132** are coupled in a local area network (LAN) **136**. In addition, a system administrator's workstation **138**, a user workstation **140**, and a supervisor's workstation **142** are coupled to LAN **136**. Alternatively, workstations **138**, **140**, and **142** are coupled to LAN **136** using an Internet link or are connected through an Intranet.

(40) Each workstation, **138**, **140**, and **142** is a personal computer having a web browser. Although the functions performed at the workstations typically are illustrated as being performed at respective workstations **138**, **140**, and **142**, such functions can be performed at one of many personal computers coupled to LAN **136**. Workstations **138**, **140**, and **142** are illustrated as being associated with separate functions only to facilitate an understanding of the different types of functions that can be performed by individuals having access to LAN **136**.

(41) Server system **112** is configured to be communicatively coupled to various individuals, including employees **144** and to third parties, e.g., account holders, customers, auditors, developers, consumers, merchants, acquirers, issuers, etc., **146** using an ISP Internet connection **148**. The communication in the example embodiment is illustrated as being performed using the Internet, however, any other wide area network (WAN) type communication can be utilized in other embodiments, i.e., the systems and processes are not limited to being practiced using the Internet. In addition, and rather than WAN **150**, local area network **136** could be used in place of WAN **150**.

(42) In the example embodiment, any authorized individual having a workstation **154** can access system **122**. At least one of the client systems includes a manager workstation **156** located at a remote location. Workstations **154** and **156** are personal computers having a web browser. Also, workstations **154** and **156** are configured to communicate with server system **112**. Furthermore, fax server **128** communicates with remotely located client systems, including a client system **156** using a telephone link. Fax server **128** is configured to communicate with other client systems **138**, **140**, and **142** as well.

(43) FIG. 4 illustrates an example configuration of a user system **202** operated by a user **201**, such as cardholder **22** (shown in FIG. 1). User system **202** may include, but is not limited to, client systems **114**, **138**, **140**, and **142**, POS terminal **118**, workstation **154**, and manager workstation **156**. In the example embodiment, user system **202** includes a processor **205** for executing instructions. In some embodiments, executable instructions are stored in a memory area **210**. Processor **205** may include one or more processing units, for example, a multi-core configuration. Memory area **210** is any device allowing information such as executable instructions and/or written works to be stored and retrieved. Memory area **210** may include one or more computer readable media.

(44) User system **202** also includes at least one media output component **215** for presenting information to user **201**. Media output component **215** is any component capable of conveying information to user **201**. In some embodiments, media output component **215** includes an output adapter such as a video adapter and/or an audio adapter. An output adapter is operatively coupled to processor **205** and operatively couplable to an output device such as a display device, a liquid crystal display (LCD), organic light emitting diode (OLED) display, or "electronic ink" display, or an audio output device, a speaker or headphones.

(45) In some embodiments, user system **202** includes an input device **220** for receiving input from user **201**. Input device **220** may include, for example, a keyboard, a pointing device, a mouse, a stylus, a touch sensitive panel, a touch pad, a touch screen, a gyroscope, an accelerometer, a position detector, or an audio input device. A single component such as a touch screen may function as both an output device of media output component **215** and input device **220**. User system **202** may also include a communication interface **225**, which is communicatively couplable

to a remote device such as server system **112**. Communication interface **225** may include, for example, a wired or wireless network adapter or a wireless data transceiver for use with a mobile phone network, Global System for Mobile communications (GSM), 3G, or other mobile data network or Worldwide Interoperability for Microwave Access (WIMAX).

(46) Stored in memory area **210** are, for example, computer readable instructions for providing a user interface to user **201** via media output component **215** and, optionally, receiving and processing input from input device **220**. A user interface may include, among other possibilities, a web browser and client application. Web browsers enable users, such as user **201**, to display and interact with media and other information typically embedded on a web page or a website from server system **112**. A client application allows user **201** to interact with a server application from server system **112**.

(47) FIG. 5 illustrates an example configuration of a server system **301** such as server system **112** (shown in FIGS. 2 and 3). Server system **301** may include, but is not limited to, database server **116**, transaction server **124**, web server **126**, fax server **128**, directory server **130**, and mail server **132**.

(48) Server system **301** includes a processor **305** for executing instructions. Instructions may be stored in a memory area **310**, for example. Processor **305** may include one or more processing units (e.g., in a multi-core configuration) for executing instructions. The instructions may be executed within a variety of different operating systems on the server system **301**, such as UNIX, LINUX, Microsoft Windows®, etc. It should also be appreciated that upon initiation of a computer-based method, various instructions may be executed during initialization. Some operations may be required in order to perform one or more processes described herein, while other operations may be more general and/or specific to a particular programming language (e.g., C, C#, C++, Java, or other suitable programming languages, etc.).

(49) Processor **305** is operatively coupled to a communication interface **315** such that server system **301** is capable of communicating with a remote device such as a user system or another server system **301**. For example, communication interface **315** may receive requests from client system **114** via the Internet, as illustrated in FIGS. 2 and 3.

(50) Processor **305** may also be operatively coupled to a storage device **134**. Storage device **134** is any computer-operated hardware suitable for storing and/or retrieving data. In some embodiments, storage device **134** is integrated in server system **301**. For example, server system **301** may include one or more hard disk drives as storage device **134**. In other embodiments, storage device **134** is external to server system **301** and may be accessed by a plurality of server systems **301**. For example, storage device **134** may include multiple storage units such as hard disks or solid state disks in a redundant array of inexpensive disks (RAID) configuration. Storage device **134** may include a storage area network (SAN) and/or a network attached storage (NAS) system.

(51) In some embodiments, processor **305** is operatively coupled to storage device **134** via a storage interface **320**. Storage interface **320** is any component capable of providing processor **305** with access to storage device **134**. Storage interface **320** may include, for example, an Advanced Technology Attachment (ATA) adapter, a Serial ATA (SATA) adapter, a Small Computer System Interface (SCSI) adapter, a RAID controller, a SAN adapter, a network adapter, and/or any component providing processor **305** with access to storage device **134**.

(52) FIG. 6 illustrates a component view **500** of the payment and record system **100** as illustrated in FIG. 2. In the example embodiment, Server system **112** includes a plurality of components for performing the functions described herein. More specifically, server system **112** comprises a processor **502**, a receiver component **504**, a transmitter component **506**, a payment network interface component **508**, a memory interface component **510**, a verification component **514**, an available account component **516**, an available personal record component **518**, a transaction reward component **520**, and an associated personal record transaction component **522**.

(53) Processor **502** may be any device capable of executing computer-executable instructions.

Processor **502** may be configured to process and direct data from one component to another component. In the example embodiment, processor **502** may be the Central Processing Unit (CPU) of server system **112**.

(54) Receiver component **504** may be any device capable of receiving a signal. In the example embodiment, receiver component **504** includes a network adapter for receiving data through a network connection such as a WAN or LAN. Receiver component **504** is communicatively coupled to at least one of client system **114**, payment processing computing device **36**, and processor **502**. Receiver component **504** is configured to receive data from payment processing computing device **36** and route the data to processor **502**.

(55) Transmitter component **506** may be any device capable of transmitting a signal. In the example embodiment, transmitter component **506** is a network adapter capable of transmitting data through a network connection such as a WAN or LAN. Transmitter component **506** is communicatively coupled to processor **502** and payment processing computing device **36**, and is configured to transmit signals from processor **502** to payment processing computing device **36**.

(56) Payment network interface component **508** may be any device or set of computer-executable instructions that enables server system **112** to communicate with interchange network **28**. In the example embodiment, payment network interface component **508** is a network adapter and sends and receives data through a network connection such as a WAN or LAN.

(57) Memory interface component **510** may be any device or set of computer-executable instructions for communicating with memory **120**. In the example embodiment, memory interface **510** communicates with memory **120** to store and retrieve data.

(58) Verification component **514** may be a device or set of computer-executable instructions for verifying the identity of at least one of cardholder **22** and merchant **24**. Verification component **514** may use, for example, a personal identification number (PIN) entered by cardholder **22** or merchant **24**. Additionally, a challenge question using transaction data associated with cardholder **22** may be presented to cardholder **22** and cardholder **22** be prompted for an answer. Because the challenge question or questions are based on cardholder **22** own transaction data, the probability that only cardholder **22** would be able to answer the challenge question(s) is relatively high.

(59) Available account component **516** may be a device or set of computer-executable instructions for determining payment accounts associated with cardholder **22** that are available for making a payment transaction. The available account component **516** may also provide a list of available accounts to cardholder computing device **121**.

(60) Available personal record component **518** may be any device or set of computer-executable instructions for determining personal records that are available to be stored or are available to be transferred. The personal records available to be transferred may be stored in memory **120**. The personal records to be stored may be located on client system **114** or cardholder computing device **121**.

(61) Transaction reward component **520** may be a device or set of computer-executable instructions for determining whether any discounts, coupons, rewards points, product reviews, or other incentives, are associated with a particular payment transaction.

(62) Associated personal record transaction component **522** may be any device or set of computer-executable instructions for determining whether an additional personal record transaction associated with at least one of a completed payment transaction and a completed personal record transaction is available.

(63) As used herein, the terms “software” and “firmware” are interchangeable, and include any computer program stored in memory for execution by processors **40**, **205**, and/or **305**. Memory area **310** may include, but are not limited to, random access memory (RAM) such as dynamic RAM (DRAM) or static RAM (SRAM), read-only memory (ROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), and non-volatile RAM (NVRAM). The above memory types are examples only, and are thus not limiting as

to the types of memory usable for storage of a computer program.

(64) The term processor, as used herein, refers to central processing units, microprocessors, microcontrollers, reduced instruction set circuits (RISC), application specific integrated circuits (ASIC), logic circuits, and any other circuit or processor capable of executing the functions described herein.

(65) As will be appreciated based on the foregoing specification, the above-discussed embodiments of the disclosure may be implemented using computer programming or engineering techniques including computer software, firmware, hardware or any combination or subset thereof. Any such resulting program, having computer-readable and/or computer-executable instructions, may be embodied or provided within one or more computer-readable media, thereby making a computer program product, i.e., an article of manufacture, according to the discussed embodiments of the disclosure. The computer readable media may be, for instance, a fixed (hard) drive, diskette, optical disk, magnetic tape, semiconductor memory such as read-only memory (ROM) or flash memory, etc., or any transmitting/receiving medium such as the Internet or other communication network or link. The article of manufacture containing the computer code may be made and/or used by executing the instructions directly from one medium, by copying the code from one medium to another medium, or by transmitting the code over a network.

(66) The above-described embodiments of a method and system of a mobile remote payment system provides a cost-effective and reliable means for receiving a consumer initiated financial payment request during a financial transaction with a merchant that may not have a POS device. More specifically, the methods and systems described herein facilitate acquiring the data needed by a consumer to submit a payment authorization request message to an interchange network. In addition, the above-described methods and systems facilitate transmitting a payment authorization request message to an issuer bank for authorization. As a result, the methods and systems described herein facilitate expanding consumers' ability to engage in commerce in a cost-effective and reliable manner.

(67) This written description uses examples to disclose the disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A payment processing computing device for processing a transaction between a consumer and a merchant, the payment processing computing device comprising a processor communicatively coupled to a consumer mobile device via an electronic communications network and communicatively coupled to a memory device, wherein the processor is programmed to: prior to generating an authorization request message, transmit, to the consumer mobile device, a challenge question associated with one or more previous transactions initiated by the consumer; prior to generating the authorization request message, receive, from the consumer mobile device and without use of one or more computing devices associated with the merchant, a consumer alias, transaction amount data, consumer verification data, and merchant data associated with the merchant, the transaction amount data comprising an image file of one or more amounts for one or more items in the transaction captured by an image capture component of the consumer mobile device and without the use of the one or more computing devices associated with the merchant, the consumer verification data comprising an answer to the challenge question, the merchant data

comprising an image of optical merchant identification information captured by the image capture component and without the use of the one or more computing devices associated with the merchant, and a geolocation of the merchant captured by a geolocation component of the consumer mobile device, wherein the image includes unencoded business-identifying information of the merchant; prior to generating the authorization request message, determine, based on at least one of a computing power of the processor or a bandwidth of the electronic communications network, to process the received image file by the processor or transmit the received image file to an interchange network for processing, wherein the interchange network is communicatively coupled to the processor via the electronic communications network; prior to generating the authorization request message and in response to determining to process the received image file by the processor, determine a merchant identifier of the merchant based on the received image and the received geolocation, and a transaction amount based on processing the image file included in the transaction amount data; verify, using merchant verification data previously received from registered merchants, that the determined merchant identifier and the merchant are associated with a registered merchant of the registered merchants; retrieve, from the memory device, information identifying an acquiring bank associated with the determined merchant identifier, wherein the registered merchant has an established account with the acquiring bank; retrieve, from the memory device, an account number based on the consumer alias and the consumer verification data, the account number associated with the consumer; generate, without receiving any information from the one or more computing devices associated with the registered merchant, the authorization request message including the account number, the determined transaction amount, and the determined merchant identifier; transmit the authorization request message over a payment network to an issuer bank associated with the consumer, wherein the registered merchant has no relationship with the issuer bank; transmit the authorization request message to the acquirer bank associated with the registered merchant; and settle the transaction between the issuer bank and the registered merchant via the acquirer bank.

2. The payment processing computing device of claim 1, wherein the payment processing computing device is further communicatively coupled to the one or more computing devices associated with the merchant and the processor is further programmed to: receive an authorization response message from the issuer bank through the payment network; and transmit the authorization response message to the consumer mobile device and the one or more computing devices associated with the merchant, wherein the one or more computing devices associated with the merchant is one of a feature phone, a smartphone, and a tablet.

3. The payment processing computing device of claim 2, wherein the consumer mobile device and the one or more computing devices associated with the merchant are wireless devices communicatively couplable to the payment processing computing device through a wireless portion of the electronic communications network.

4. The payment processing computing device of claim 1, wherein the consumer verification data includes a personal identification number (PIN) of the consumer, and the merchant verification data includes a personal identification number (PIN) of the registered merchant.

5. The payment processing computing device of claim 1, wherein the processor is further programmed to generate the challenge question using transaction data associated with the consumer.

6. The payment processing computing device of claim 1, wherein the consumer alias includes one of a phone number and an email address of the consumer.

7. The payment processing computing device of claim 1, wherein the consumer alias includes a device identifier associated with the consumer mobile device.

8. A computer-implemented method for processing a transaction between a merchant and a consumer, the method implemented using a payment processing computing device including a processor communicatively coupled to a consumer mobile device via an electronic communications

network and communicatively coupled to a memory device, the method comprising steps performed by the processor of: prior to generating an authorization request message, transmitting, to the consumer mobile device, a challenge question associated with one or more previous transactions initiated by the consumer; prior to generating the authorization request message, receiving, from the consumer mobile device and without use of one or more computing devices associated with the merchant, a consumer alias, transaction amount data, consumer verification data, and merchant data associated with the merchant, the transaction amount data including an image file of one or more amounts for one or more items in the transaction captured by an image capture component of the consumer mobile device and without the use of the one or more computing devices associated with the merchant, the consumer verification data including an answer to the challenge question, the merchant data including an image of optical merchant identification information captured by the image capture component and without the use of the one or more computing devices associated with the merchant, and a geolocation of the merchant captured by a geolocation component of the consumer mobile device, wherein the image includes unencoded business-identifying information of the merchant; prior to generating the authorization request message, determine, based on at least one of a computing power of the processor or a bandwidth of the electronic communications network, to process the received image file by the processor or transmit the received image file to an interchange network for processing, wherein the interchange network is communicatively coupled to the processor via the electronic communications network; prior to generating the authorization request message and in response to determining to process the received image file by the processor, determining a merchant identifier of the merchant based on the received image and the received geolocation, and a transaction amount based on processing the image file included in the transaction amount data; verifying, using merchant verification data previously received from registered merchants, that the determined merchant identifier and the merchant are associated with a registered merchant of the registered merchants; retrieving, from the memory device, information identifying an acquiring bank associated with the determined merchant identifier, wherein the registered merchant has an established account with the acquiring bank; retrieving, from the memory device, an account number based on the consumer alias and the consumer verification data, the account number associated with the consumer; generating, without receiving any information from the one or more computing devices associated with the registered merchant, the authorization request message including the account number, the determined transaction amount, and the determined merchant identifier; transmitting the authorization request message over a payment network to an issuer bank associated with the consumer, wherein the registered merchant has no relationship with the issuer bank; transmitting the authorization request message to the acquirer bank associated with the registered merchant; and settling the transaction between the issuer bank and the registered merchant via the acquirer bank.

9. The method of claim 8, wherein the payment processing computing device is further communicatively coupled to the one or more computing devices associated with the merchant, and wherein the method further comprises: receiving an authorization response message from the issuer bank through the payment network; and transmitting the authorization response message to the consumer mobile device and the one or more computing devices associated with the merchant, wherein the one or more merchant devices is one of a feature phone, a smartphone, and a tablet.

10. The method of claim 9, wherein the consumer mobile device and the one or more computing devices associated with the merchant are wireless devices communicatively coupleable to the payment processing computing device through a wireless portion of the electronic communications network.

11. The method of claim 8, wherein the consumer verification data includes a personal identification number (PIN) of the consumer, and the merchant verification data includes a personal identification number (PIN) of the registered merchant.

12. The method of claim 8 further comprising generating the challenge question using transaction data associated with the consumer.
13. The method of claim 8, wherein the consumer alias includes one of a phone number and an email address of the consumer.
14. The method of claim 8, wherein the consumer alias includes a device identifier associated with the consumer mobile device.
15. At least one non-transitory computer-readable storage medium having computer-executable instructions embodied thereon, wherein when executed by a payment processing computing device for processing a transaction between a consumer and a merchant, the payment processing computing device including a processor communicatively coupled to a consumer mobile device via an electronic communications network and communicatively coupled to a memory device, the computer-executable instructions cause the processor to: prior to generating an authorization request message, transmit, to the consumer mobile device, a challenge question associated with one or more previous transactions initiated by the consumer; prior to generating the authorization request message, receive, from the consumer mobile device and without use of one or more computing devices associated with the merchant, a consumer alias, transaction amount data, consumer verification data, and merchant data associated with the merchant, the transaction amount data including an image file of one or more amounts for one or more items in the transaction captured by an image capture component of the consumer mobile device and without the use of the one or more computing devices associated with the merchant, the consumer verification data including an answer to the challenge question, the merchant data including an image of optical merchant identification information captured by the image capture component and without the use of the one or more computing devices associated with the merchant and a geolocation of the merchant captured by a geolocation component of the consumer mobile device, wherein the image includes unencoded business-identifying information of the merchant; prior to generating the authorization request message, determine, based on at least one of a computing power of the processor or a bandwidth of the electronic communications network, to process the received image file by the processor or transmit the received image file to an interchange network for processing, wherein the interchange network is communicatively coupled to the processor via the electronic communications network; prior to generating the authorization request message and in response to determining to process the received image file by the processor, determine a merchant identifier of the merchant based on the received image and the received geolocation, and a transaction amount based on processing the image file included in the transaction amount data; verify, using merchant verification data previously received from registered merchants, that the determined merchant identifier and the merchant are associated with a registered merchant of the registered merchants; retrieve, from the memory device, information identifying an acquiring bank associated with the determined merchant identifier, wherein the registered merchant has an established account with the acquiring bank; retrieve, from the memory device, an account number based on the consumer alias and the consumer verification data, the account number associated with a consumer; generate, without receiving any information from the one or more computing devices associated with the registered merchant, the authorization request message including the account number, the determined transaction amount, and the determined merchant identifier; transmit the authorization request message over a payment network to an issuer bank associated with the consumer, wherein the registered merchant has no relationship with the issuer bank; transmit the authorization request message to the acquirer bank associated with the registered merchant; and settle the transaction between the issuer bank and the registered merchant via the acquirer bank.
16. The computer-readable storage medium of claim 15, wherein the payment processing computing device is further communicatively coupled to the one or more computing devices associated with the merchant, and wherein the computer-executable instructions further cause the processor to: receive an authorization response message from the issuer bank through the payment

network; and transmit the authorization response message to the consumer mobile device and the one or more computing devices associated with the merchant, wherein the one or more computing devices associated with the merchant is one of a feature phone, a smartphone, and a tablet.

17. The computer-readable storage medium of claim 15, wherein the consumer verification data includes a personal identification number (PIN) of the consumer, and the merchant verification data includes a personal identification number (PIN) of the registered merchant.

18. The computer-readable storage medium of claim 15, wherein the computer-executable instructions further cause the processor to generate the challenge question using transaction data associated with the consumer.

19. The computer-readable storage medium of claim 15, wherein the consumer alias includes one of a phone number and an email address of the consumer.

20. The computer-readable storage medium of claim 15, wherein the consumer alias includes a device identifier associated with the consumer mobile device.
